

The exponential growth of robotics in India is fueling the demand for skilled workforce that can design, build, operate and repair robots and robotic machines.

—Anuj Kapuria, CEO, Hi Tech Robotic Systemz



cue operations, robots will touch our lives in several ways.

Further applying artificial intelligence and robotics, one can create unmanned aerial vehicles for weather prediction, androids for assistance and smart munitions for defence applications.

“One can explore career opportunities in diversified sectors ranging from defence research to pure manufacturing. The exponential growth of robotics in India is fueling the demand for skilled workforce that can design, build, operate and repair robots and robotic machines,” says Anuj Kapuria, CEO, Hi Tech Robotic Systemz.

Find your role

I believe, up to this point, you have grabbed enough information about the opportunities-to-be in the robotics field. As electronics professional, you may find placement in the design section of a robot manufacturing firm. “Fresh electronics graduates can definitely play important roles in some of the positions, especially in the creative designing and problem-solving sections. They may have to work on electronic designs and embedded programs for robotic assemblies like actuators, sensors and interface circuitries and develop innovative GUI interfaces and control mechanisms for the robots,” opined Pulakit Gaur, CEO and co-founder of Gridbots.

In the manufacturing section, the applicants may play important roles in maintenance and operation. Microchip manufacturers, like Intel, recruit robotics and artificial intelligence specialists. Employment opportunities also exist in education and research domains.

Electronics departments in most of the IITs and CSIR labs offer research fellowships in robotics and artificial intelligence.

As a robotics engineer, you may even grab employment opportunities abroad in robot programming, troubleshooting and maintenance. Fellowships are also available in research institutes, emphasising on humanoid robotics and computational neuroscience. Prestigious research organisations like NASA may be your ultimate destination.

The next question should be how to get an entry ticket? The answer is: Only a decent technical degree (diploma or graduation) is not enough to get you a job in robotics. You need to have a knack for the field—be it in the form of a project or training. Recruiters also look for specialised skills in research and development areas in a postgraduate or doctorate applicant. Most of the employers take interns and train them directly in various areas in robotics.

Thankfully, recruiters in this sector believe in talent building. For example, Kapuria takes talent building as part of his social responsibility. He enthusiastically talks about the involvement of interns in live projects who were later absorbed by the company.

There are opportunities for diploma holders as well, but the situation is a little bit daunting. Initially, they may have to be restricted within the fields of PCB and mechanical designing, hardware integration and soldering.

Money matters

Let's check whether the robotics field supports the feeling that being fairly rewarded for your work is a substantial part of career satisfaction.

Students who have completed

their postgraduation and doctorate are absorbed at Rs 450,000 to 950,000 per annum. Those who are graduating may get offers ranging from Rs 350,000 to 650,000 per annum.

“A robotics engineer could be offered a minimum of Rs 350,000 per annum. The growth rate, however, is quite steep as the industry offers lots of prospects,” says Kapuria. He also adds that it is not the experience but the actual expertise like problem-solving skills specific to robotics domains which determine the career growth.

The field also offers a decent starting salary in the range of Rs 10,000 to Rs 15,000 per month to diploma holders, although the scope of work is somewhat restricted for them.

How to begin?

Possibly, you will never come across any dedicated session on robotics throughout your curriculum. Don't get disappointed. Let us take the experts' help. Experts like Dr Prahlad Vadakkepatt, director, Robhatah Robotic Solutions and a faculty member of National University of Singapore, say that the basic knowhow of electronics as well as other allied fields like electrical engineering and mechanical engineering are quite important. But it is no longer enough to know only your fundamentals. The experts also specifically explain that the professionals work in a cut-throat environment and they should know survival tactics in order to compete in the field.

There are opportunities in the industrial robotics space focusing on developments specific to Indian requirements or expansions along manufacturing. The mobile robotics space is relatively new, and may offer higher potential growth in the coming decades, where a variety of applications from consumer products to medical robotics are expected to offer growth opportunities.

From an industrial robotics perspective, higher education in specialised areas like kinematics, statics, dynamics, path planning, control,